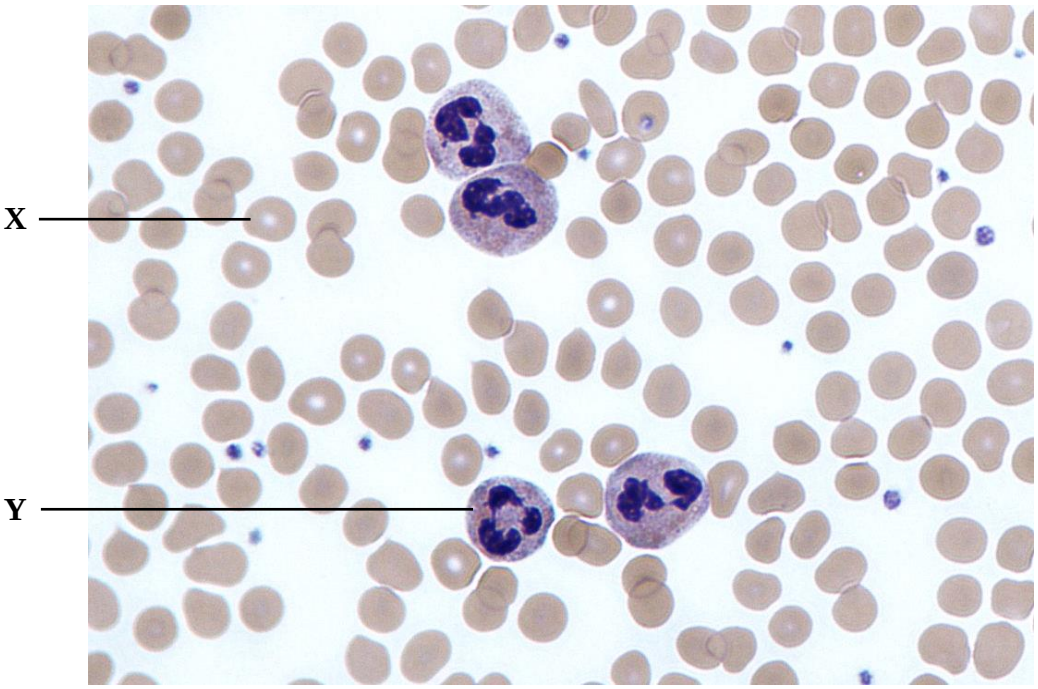
Structured questions

|!|BCELA0250000001|!| ★★

A scientist vaccinated a man with a type of live but weakened virus. Immediately after vaccination, the scientist collected a blood sample from the man. The photomicrograph below shows the blood smear of the sample collected immediately after vaccination.



Human blood smear immediately after vaccination, W.M. (×1200)

(a) Suggest ***two*** reasons why the virus injected are ***not*** visible in the photomicrograph. (2 marks)

(b) The diameter of cell Y in the photomicrograph is 1 cm. Calculate its actual diameter. Show your workings. (2 marks)

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(c) (i) Five days later, the scientist collected another blood sample from the man for analysis. Predict the change in the number of cell type Y with 5 days before. (1 mark)

(ii) What is the significance of this change? (2 marks)

(d) Blood can also be used to detect genetic diseases. Suggest and explain which cell type, X or Y, should be used. (2 marks)

##

(a) Any ***two*** of the following: (max. 2)

- The viruses are too small to be seen under this magnification. (1)
- The viruses may not be present in the blood smear shown because only a small amount of blood is used to produce a smear. (1)
- The virus causes regional or local infection in tissues other than blood, therefore cannot be found in the blood. (1)

(b) Actual diameter of cell Y = 10000 μm ÷ 1200 (1)

= 8.33 μm (1)

(c) (i) The number of cell type Y increases. (1)

(ii) Cell type Y can engulf pathogens and digest them with enzymes. (1)

Increasing the number of cell type Y can help the body destroy the invading pathogens faster. (1)

(d) Cell type Y (1)

It contains a complete set of chromosomes / genetic material. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Mathematical skills	2
(c)(i)	Applying concepts	1
(c)(ii)	Understanding basic concepts	2
(d)	Applying concepts	2

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|!|BCELB0250000002|!| ★ **Cross topic** **STSE**

There were outbreaks of rubella (德國麻疹) in Japan in 2018. Read the information below about rubella and answer the questions that follow:

> Rubella is a highly infectious disease caused by rubella virus. People infected with rubella usually have a rash, fever, headache, enlargement of lymph nodes, upper respiratory symptoms and conjunctivitis. Rubella infection may also cause birth defects in the developing foetus. Deafness, cataract, heart malformations, mental retardation etc., is likely to occur in infants born to women who got infected during the first three months of pregnancy.

(a) Explain why rubella cannot be treated with antibiotics. (2 marks)

(b) The Centre for Health Protection (CHP) advises travellers who will be visiting Japan to receive vaccination against rubella before their trip. Using your biological knowledge, explain how vaccination helps prevent rubella infection. (4 marks)

(c) The CHP also appeals to pregnant women and women preparing for pregnancy who are non-immune to rubella to avoid travelling to Japan during the outbreak period, because pregnant women should not receive rubella-containing vaccine.

(i) Suggest a reason why pregnant women should not receive rubella-containing vaccine. (2 marks)

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(ii) Pregnant women who are immune to rubella can pass the immunity against rubella to their foetus. Suggest how. (2 marks)

(iii) Name the type of immunity you answered in (ii). (1 mark)

##

(a) Antibiotics work by interfering vital cellular processes. (1)
Viruses are non-cellular entities and have no cellular activities. (1)

(b) Vaccines contain rubella virus (1)
which stimulates the immune system to produce memory cells for the rubella virus. (1)
When the same rubella virus invades the vaccinated person again, (1)
the memory cells produce large amounts of antibodies quickly to eliminate the invading rubella virus. (1)

(c) (i) The rubella virus in the vaccine is small enough to pass through the placenta (1)
and cause harm to the developing foetus. (1)
(ii) Antibodies against rubella virus present in the pregnant women's blood (1)
can cross the placenta and enter the foetus' bloodstream. (1)
(iii) Passive immunity (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Understanding basic concepts	4
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	2
(c)(iii)	Understanding basic concepts	1

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|!|BCELB0250000003|!| ★ **Cross topic**

Hepatitis E is a liver disease caused by the hepatitis E virus. The virus is usually transmitted by food or water that has been contaminated with the faeces of an infected person. Three persons (P, Q and R) took blood tests. From the results of the blood tests, they can check whether they are immune to hepatitis E.

	P	**Q**	**R**
Hepatitis E antigen in the blood	−	−	+
Antibody against hepatitis E antigen in the blood	++	−	+++

(Remarks: number of '+' represents the amount of the substance in the blood; '-' represents the absence of the substance in the blood)

(a) Based on the results, identify which person(s) is/are immune to hepatitis E. (1 mark)

_____

(b) The three persons have never received hepatitis E vaccination. Suggest explanation for the difference in their blood tests. (3 marks)

_____

_____

_____

_____

(c) Hepatitis A is caused by the hepatitis A virus. Symptoms of the disease are similar to those of hepatitis E. Explain whether hepatitis E vaccination can provide protection against hepatitis A. (2 marks)

_____

_____

_____

_____

(d) In 2018, there were two reported cases of patients infected with the rat hepatitis E virus. The rat hepatitis E virus may have been transmitted to humans by rats. According to this possible transmission route, suggest ***two*** measures to help prevent the spread of the rat hepatitis E virus. (2 marks)

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##

(a) P, R (1)

(b) Persons P and R have contracted hepatitis E before. (1)

Their immune system has produced antibodies to destroy the virus. (1)

Q has not contracted hepatitis E before. (1)

(c) No, (1)

because the immunological memory of the white blood cells is specific to hepatitis E virus but not to hepatitis A virus. (1)

(d) Any ***two*** rat control measures, e.g. (2)

- Use traps and poison to catch and kill rats. (1)
- Eliminate shelter for rats, e.g. clear away any refuse and disused articles at homes and in the communal areas of a building / regularly clear up the void underneath or behind bulky furniture / inspect any concealed places like false ceilings, air ducts and wire ducts / seal any holes or crevices in the ceilings, walls or floors / keep flowerbeds clean and tidy without accumulation of refuse and articles. (1)
- Eliminate any potential food sources for rats, e.g. clear all rubbish piles and refuse / ensure that all garbage containers and dumpsters have a tight fitting cover / avoid keeping surplus food in bird feeders and pet bowls. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	3
(c)	Understanding basic concepts	2
(d)	Applying concepts	2

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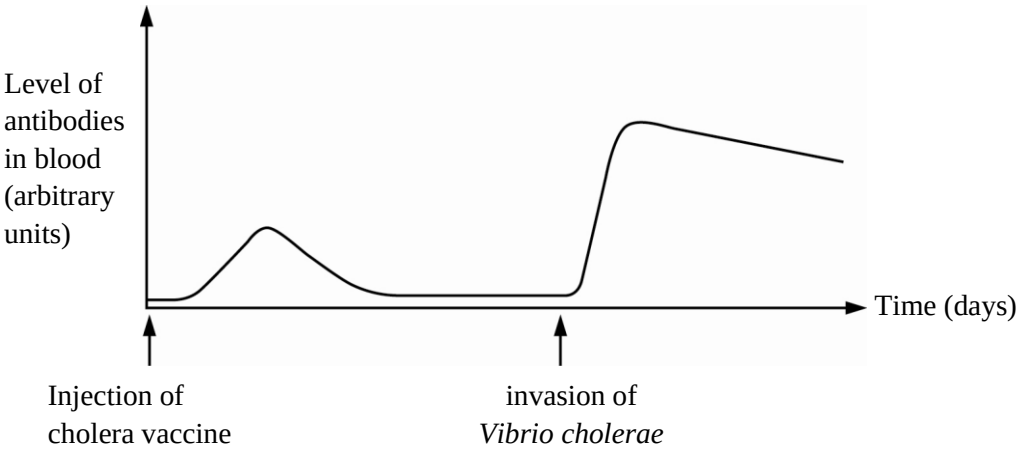
|!|BCELB0250000004|!| ★

Cholera is an acute intestinal infection caused by the bacterium *Vibrio cholerae*. The bacteria infect the small intestine and produce toxin which leads to watery diarrhoea. Cholera can be fatal in severe cases. Travellers may get vaccination against cholera if they are going to areas with cholera outbreak.

(a) State the mode of transmission of cholera. (1 mark)

(b) Suggest why areas with cholera outbreaks are usually within underdeveloped countries. (1 mark)

(c) The graph below shows the changes in the level of antibodies in blood following an injection of a cholera vaccine and then an invasion by the bacterium *Vibrio cholerae*.



(i) What is a vaccine? (1 mark)

(ii) Explain why the level of antibodies does not increase immediately after the injection of the vaccine. (1 mark)

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(iii) State ***two*** differences between the changes in antibody levels induced by the vaccine injection and the bacterial invasion. Explain the differences. (5 marks)

(d) Apart from vaccination, suggest another way by which people can prevent cholera. (1 mark)

##

(a) Through the ingestion of contaminated food or water (1)

(b) Underdeveloped countries usually have poor hygiene / poor sewage treatment / poor personal health / poor public health. (1)

(c) (i) A vaccine is a dose of weakened or killed pathogens or its antigens. (1)

(ii) It takes time for lymphocytes to come into contact with the antigens and to produce antibodies. (1)

(iii) Any ***two*** of the following differences: (2)

	Vaccine injection	**Bacterial invasion**
Rise in the antibody level	slower	faster
Peak of the antibody level	lower	higher
Latent period	longer	shorter

Explanation:

After the injection of the vaccine, memory cells for the antigen of the bacteria have been developed. (1)

When the same bacteria enter the body the second time, the memory cells quickly differentiated into cells that produce a large amount of antibodies in a short period of time. (1)

(d) Any ***one*** of the following: (1)

- Cook food thoroughly
- Drink boiled water

(Or other reasonable answer)

(10 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
(c)(iii)	Analysing data	2
	Applying concepts	3
(d)	Understanding basic concepts	1

|!|BCELC0250000005|!|

Tom broke a beaker and cut his finger during an experiment. He was instructed to wash the wound with disinfectants and wrap the wound with a plaster.

(a) Explain why bleeding at the wound stopped after a while. (2 marks)

(b) State the changes in the following structures at the wound.

(i) arterioles (1 mark)

(ii) capillary walls (1 mark)

(c) Describe how the two changes mentioned in (b) facilitated phagocytosis at the wound. (3 marks)

(d) Pathogens cannot enter the body if the skin remains intact.

(i) Name ***one*** secretion of the skin which kills pathogens. (1 mark)

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(ii) Which part of the skin produces the secretion? (1 mark)

##

(a) Fibrinogen in the blood was converted to fibrin which formed a network, trapping blood platelets and blood cells. (1)

This formed a blood clot which stopped the bleeding. (1)

(b) (i) The arterioles dilated. (1)

(ii) The capillary walls became more permeable. (1)

(c) More phagocytes in the blood were brought to the wound when the arterioles dilated. (1)

More phagocytes are forced into the wounded area when the permeability of capillary walls increased. (1)

Therefore, more phagocytes could engulf and destroy the pathogens in the wound. (1)

(d) (i) sebum (1)

(ii) sebaceous gland (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)	Understanding basic concepts	3
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	1

|!|BCELB0250000006|!| ★ **STSE**

Human immunodeficiency virus (HIV) is a pathogen which causes the acquired immune deficiency syndrome (AIDS). This pathogen can weaken the human immune system, so patients will easily get infected with other diseases and may die finally. Therefore, AIDS patients are killed by HIV indirectly.

(a) The immune system is weakened as the virus invades and destroys a type of cells in the blood. State this type of cells and explain why destroying these cells can weaken the immune system. (2 marks)

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(b) When HIV invades our bodies, antibodies against HIV are produced.

(i) After HIV has entered a cell, it can be spread by direct cell-to-cell contact. Suggest why this makes the antibodies produced ineffective in eliminating the virus. (2 marks)

(ii) In some HIV carriers, the concentration of HIV in the blood is so low that the virus is undetectable. In these cases, doctors cannot determine whether a person is an HIV carrier by detecting HIV in blood. Suggest an alternative method to test for HIV carriers. (1 mark)

(c) Suggest ***one*** disease that many AIDS patients suffer from. (1 mark)

(d) Suggest ***two*** preventive measures against the spread of HIV. (2 marks)

##

(a) white blood cells (lymphocytes) / T cells / helper T cells (1)
These cells destroy invading pathogens. (1)

(b) (i) Antibodies cannot enter the infected cells (1)
to kill the virus inside / stop the virus inside from multiplying. (1)

(ii) Test for the presence of antibodies against HIV in blood (1)

(c) pneumonia / other reasonable answer (1)

(d) Any ***two*** of the following: (2)

- Wear gloves when handling wounds and body fluids
- Avoid sharing syringes, needles and other objects that may be contaminated with blood
- Screen blood before transfusion
- Practise safe sex and wear a condom during sexual contact

(Or other reasonable answers)

(8 marks)

##

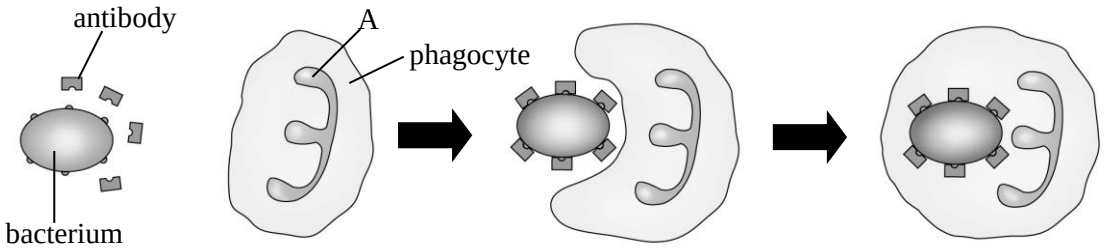
	Skills assessed	Marks
(a)	Applying concepts	2
(b)(i)	Applying concepts	2

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(b)(ii)	Applying concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	2

|!|BCELC0250000007|!|

The diagram below shows the process by which a phagocyte engulfs a bacterium with the help of antibodies.



(a) Name structure A. (1 mark)

(b) The antibodies bind to the bacterium and this enhances phagocytosis.

(i) Which type of cells produces antibodies? (1 mark)

(ii) Explain how antibodies against this bacterium are produced. (3 marks)

(iii) Antibodies act on invading pathogens in different ways. Suggest ***one*** way in which an antibody works, apart from the one shown in the diagram. (1 mark)

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(c) Describe what will happen after the bacterium is engulfed by the phagocyte.

(2 marks)

_____

_____

##

(a) nucleus (1)

(b) (i) plasma cells (1)

(ii) When B cells having the antigen receptors specific for the antigen of this bacterium binds to the antigen, they become activated. (1)

The activated B cells multiply and differentiate into plasma cells and memory B cells. (1)

Plasma cells produce antibodies specific to the antigen of this bacterium. (1)

(iii) Any ***one*** of the following: (1)

- It makes holes on the surfaces of bacteria and causes lysis of the bacteria.
- It neutralizes the toxins secreted by bacteria.
- It causes pathogens to clump together.

(c) Lysosome fuses with the vacuole which contains the bacterium. (1)

Lysosome releases lysozyme which digests the bacterium. (1)

(8 marks)

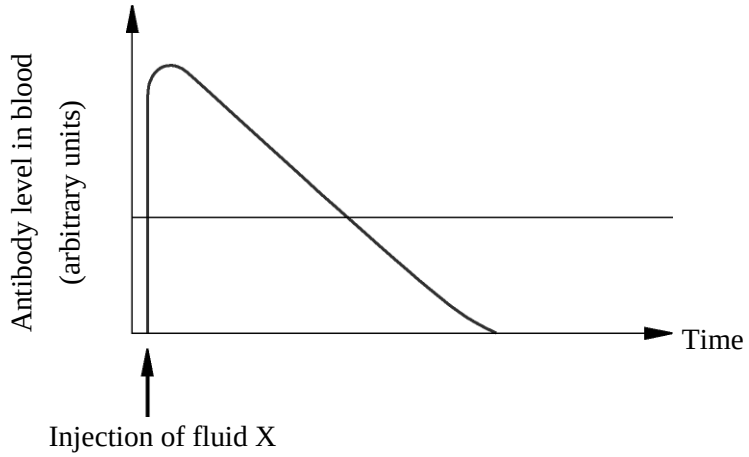
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	3
(b)(iii)	Understanding basic concepts	1
(c)	Understanding basic concepts	2

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|!|BCELC0250000008|!|

Sarah visited her family doctor because she felt sick suddenly. Her doctor gave her an injection of fluid X. After several days, Sarah recovered.
The following graph shows the changes in the antibody level in her blood after the injection of fluid X.



(a) (i) Name fluid X. (1 mark)

(ii) State the type of immunity induced by the injection of fluid X. (1 mark)

(b) State and explain

(i) the advantage of the injection of fluid X. (2 marks)

(ii) the disadvantage of the injection of fluid X. (3 marks)

(c) Sarah's friend, Sally, was injected with a vaccine against the disease suffered by Sarah. State ***one*** difference in the antibody level in the blood between Sarah and Sally who were injected with fluid X and vaccine respectively. (1 mark)

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##

(a) (i) serum (1)

(ii) passive immunity (1)

(b) (i) It provides immediate immunity (1)

because the serum contains a high concentration of the antibodies, which can help fight against the pathogens after the injection. (1)

(ii) The induced immunity cannot last long (1)

because the antibodies are not made by the immune system of the patient. (1)

The antibodies will be gradually used up or broken down. (1)

(c) Any ***one*** of the following: (1)

- The rate of the increase in the antibody level in the blood is lower in Sally.
- The maximum antibody level in the blood is lower in Sally.

(Or other reasonable answer)

(8 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	3
(c)	Understanding basic concepts	1

|!|BCELB0250000009|!| ★

Flu season is a time period during which the outbreaks of influenza become prevalent annually. Vaccination is provided by the Hong Kong government to protect citizens against the seasonal flu. Those vaccines contain antigens of the influenza virus.

(a) How does the vaccine induce immunity against influenza? (5 marks)

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(b) Suggest ***two*** preventive measures to reduce the risk of getting influenza. (2 marks)

_____

_____

(c) (i) Explain how a newborn baby may get immunity against some types of influenza from his mother. (2 marks)

_____

_____

(ii) This immunity only lasts for about six months. Why? (1 mark)

_____

##

(a) After the injection of vaccine into the blood, the antigens stimulate the B cells to multiply and differentiate into plasma cells and memory B cells. (1)
The plasma cells produce specific antibodies against the influenza virus. (1)
The antigens can be memorized by the memory B cells. (1)
Subsequent exposure to the influenza virus will initiates a rapid production of a large amount of antibodies against the virus. (1)
As the level of antibody in the blood increases to a certain level, this can prevent the development of flu. (1)

(b) Any ***two*** of the following: (2)
- wear a surgical mask when in close contact with infected persons
- avoid going to crowded and poorly ventilated public places
- clean hands with liquid soap or disinfectants frequently

(Or other reasonable answers)

(c) (i) The mother may have the antibodies against some types of influenza virus in her blood. (1)
These antibodies get across the placenta and enter the blood of the foetus during pregnancy. This provides immunity in the newborn baby. (1)

(ii) This is because the mother's antibodies will be broken down by the baby. (1)

(10 marks)

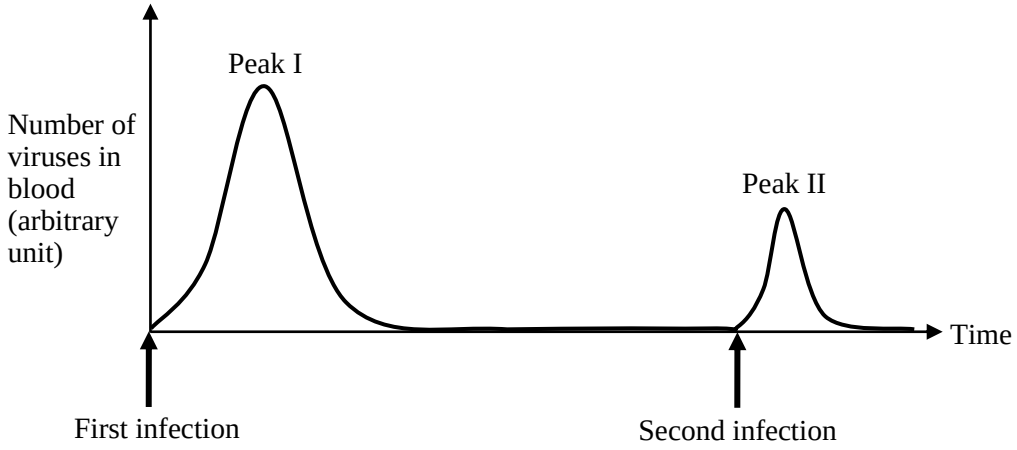
##

	Skills assessed	Marks
(a)	Applying concepts	5
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	1

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|!|BCELB0250000010|!| ★

The following graph shows the changes in the number of viruses in the blood when Wendy was infected with the same type of virus twice.



(a) Describe and explain the changes in the number of viruses in blood after the first viral infection. (5 marks)

(b) Describe and explain ***one*** difference between peak I and peak II. (5 marks)

##

(a) After Wendy was infected with the virus, the virus multiplied and its concentration increased to a maximum value. (1)
This is because initially only a few lymphocytes could recognize the antigens of the viruses. (1)
The rate of destroying the viruses was slower than that of the rate of viral multiplication. (1)
Later when the lymphocytes multiplied and produced enough antibodies to destroy the viruses, (1)
the concentration of the virus gradually decreased to zero. (1)

(b) Peak II is lower than peak I / Peak II has a steeper slope than peak I. (1)
This is because when the virus invaded the body for the first time, some B cells differentiated into memory B cells which memorize the antigen of the virus. (1)
When the same type of virus invaded the body again, the memory B cells recognized the antigens of the viruses quickly and produced a large number of plasma cells. (1)
The plasma cells produced a large amount of antibodies within a short time. (1)
This led to more viruses being destroyed per unit time / a higher rate of elimination of the viruses. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	3
(b)	Analysing data	1
	Applying concepts	4

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|!|BCELC0250000011|!|

During Physical Education (PE) lesson, a student fell over and hurt his hands and legs when he was playing basketball. He was bleeding and the wound showed inflammation.

(a) Describe how the skin can defend against the invasion of pathogens. (2 marks)

(b) Bleeding stopped as a result of blood clotting. State the importance of blood clotting. (2 marks)

(c) (i) Give ***two*** signs of inflammation at the wound. (2 marks)

(ii) What causes the signs in (i)? (2 marks)

(d) Describe how the specific immune response of the body can destroy the bacteria that have entered the tissue fluid through the wound. (3 marks)

##

(a) The dead cornified layer of the epidermis is impermeable to pathogens. (1)
Sebum secreted by the sebaceous glands in the skin contains chemicals which kill certain pathogens. (1)

(b) It reduces blood loss through the wound. (1)
It prevents pathogens from entering the body. (1)

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(c) (i) Any ***two*** of the following: (2)

- The wound is swollen.
- The wound is red.
- The wound is hot.
- The wound is painful.

(ii) Any ***two*** of the relevant causes: (2)

- The wound is swollen because more blood plasma and phagocytes are forced into the wound and tissue fluid accumulates at the wound.
- The wound is red because more blood flows to the wounded area.
- The wound is hot because more blood flows to the wounded area.
- The wound is painful because the swollen tissues press on and stimulate the pain receptors.

(d) When B cells having the antigen receptors specific for the antigen of the bacteria binds to the antigen, they become activated. (1)

The activated B cells multiply and differentiate into plasma cells and memory B cells. (1)

Plasma cells produce antibodies specific to the antigen of the bacteria. (1)

(11 marks)

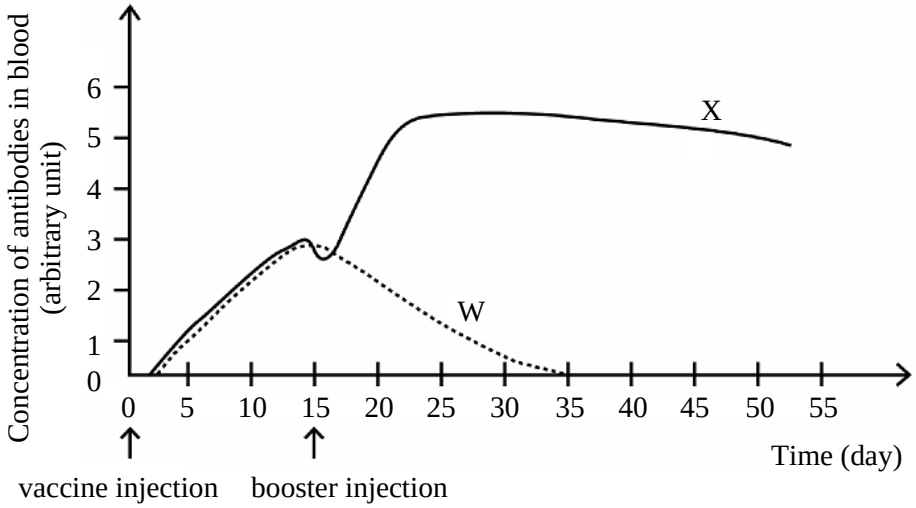
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	2
(d)	Understanding basic concepts	3

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|!|BCELB0250000012|!| ★

The graph below shows two curves: W and X.
Curve W shows the changes in the concentration of antibodies in the blood resulting from a single injection of a vaccine against a pathogen.
Curve X shows the changes in the concentration of antibodies in the blood resulting from the injections of a vaccine and then a booster against the pathogen.



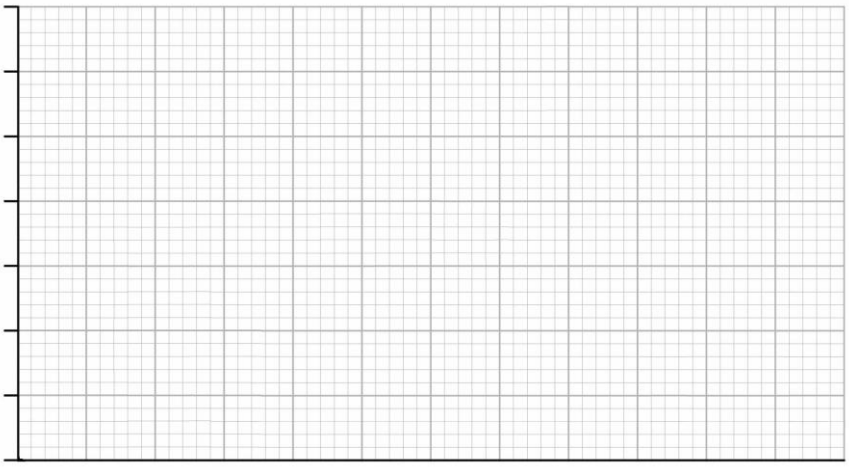
(a) Consider curve X. What is the percentage change in the maximum concentration of antibodies in the blood brought about by the injection of the vaccine and then the booster? Show your working. (2 marks)

(b) State ***two*** differences between the two curves as shown above. (2 marks)

(c) The response to the booster injection is the secondary response. Describe how the secondary response is brought about. (3 marks)

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(d) Using the same scale as the above graph, plot a graph to show the changes in the concentration of antibodies in the blood after the injection of a serum for defence against the same pathogen. (4 marks)



##

(a) Percentage change = (5.5 – 3) / 3 × 100% (1)

= + 83.3% (1)

(b) Any ***two*** of the following: (2)

W	***X***
The peak level is lower.	The peak level is higher.
The time taken to reach the peak after the injection of the vaccine is longer.	The time taken to reach the peak after the injection of the booster is shorter.
The concentration of antibodies in the blood maintains at a high level for a shorter period of time.	The concentration of antibodies in the blood maintains at a high level for a longer period of time.

(Or other reasonable answers)

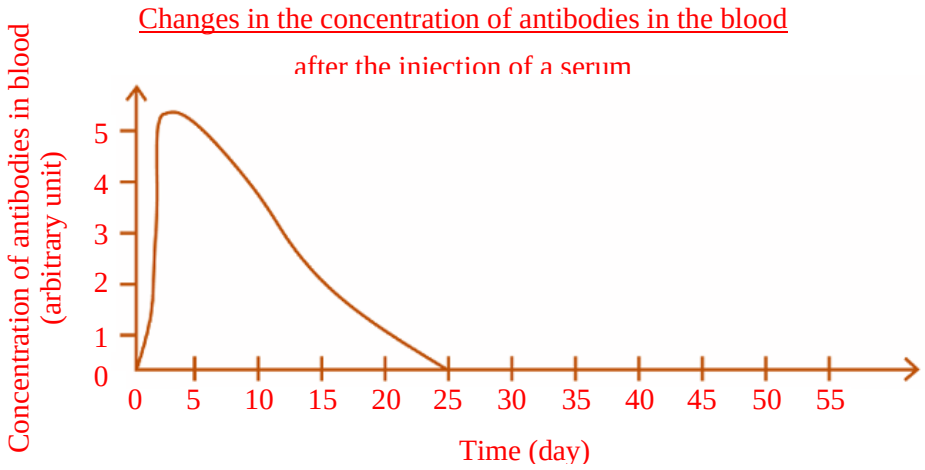
(c) In the first exposure to the antigens from the vaccine, the antigens are memorized by memory B cells. (1)

Subsequent exposure to the same antigens will result in a rapid production of a large number of plasma cells. (1)

Plasma cells produce a large amount of specific antibodies against the pathogens. (1)

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(d)



Correct title (1)
Correct choice of axes (1)
Correct labelling of axes (1)
Correct drawing of curve (1)

(11 marks)

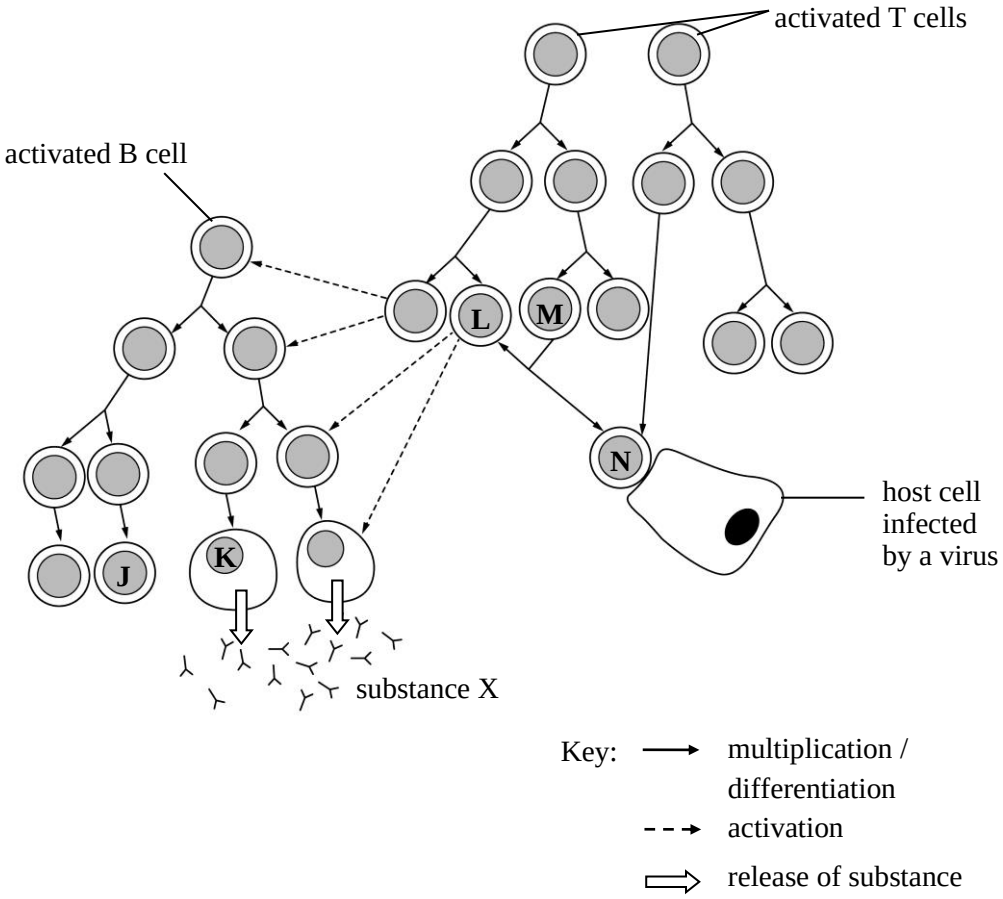
##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	2
(c)	Understanding basic concepts	3
(d)	Presenting information	4

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|!|BCELB0250000013|!| ★

The following diagram illustrates the changes that occur in B cells and T cells activated by an invading virus.



(a) From which type of cell do B cell and T cell originate? (1 mark)

(b) Identify cell K and substance X. (2 marks)

(c) (i) Describe the functions of cell M. (2 marks)

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(ii) Using the letter in the diagram, state ***one*** cell that has similar functions as cell M. (1 mark)

(d) State how cells L and N help to fight against viral infection. (4 marks)

##

(a) stem cell in the bone marrow (1)

(b) Cell K: plasma cell (1)

Substance X: antibody (1)

(c) (i) Cell M (memory T cell) 'memorsizes' the antigens of the pathogens when the pathogens first invade the body. (1)

When the same pathogen invades the body again, cell M differentiates into killer T cell quickly to destroy the cells infected by the pathogen. (1)

(ii) Cell J (1)

(d) Cell L (helper T cell) activates B cells to become plasma cells which produce antibodies against the invading viruses. (1)

Cell L also activates T cells to differentiate into cell N (killer T cell). (1)

Cell N (killer T cell) binds to the antigens on the surfaces of cells infected by the virus (1)

and directly destroys the infected cells. (1)

(10 marks)

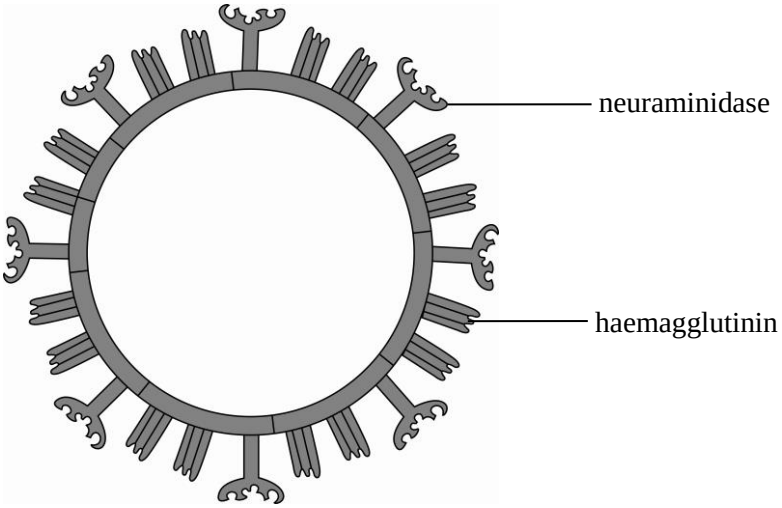
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Analysing data	1
(d)	Understanding basic concepts	4

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|!|BCELA0250000014|!| ★★

The diagram below shows some of the structures on the surface of an influenza virus.



Haemagglutinin and neuraminidase are protein molecules found on the surface of influenza virus. The names of the subtypes of influenza viruses are determined by the types of these proteins on their surface.

Haemagglutinin binds to receptor molecules on the surface of target cells, such as cells in the upper respiratory tract.

Neuraminidase is an enzyme which breaks the binding of the viruses to host cells so that the newly synthesized viruses in the host cells can be released from the cells.

(a) Explain why there are different subtypes of influenza virus. (1 mark)

(b) (i) Describe how the cells infected by the influenza virus are destroyed by T cells. (3 marks)

(ii) Describe how B cells act against the influenza virus. (3 marks)

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(c) One type of drug that is used to treat influenza is a neuraminidase inhibitor. Suggest how this type of drug could treat influenza. (2 marks)

##

(a) There are genetic variation between different types of influenza virus, which affects the types of haemagglutinin and neuraminidase molecules present on the surfaces of different types of influenza virus. (1)

(b) (i) Helper T cells recognize the antigens present on the surface of the cells infected by the influenza virus. (1)
Helper T cells then differentiate into killer T cells. (1)
Killer T cell binds to the antigen on the surface of the infected cell and destroys the cells directly. (1)

(ii) B cells differentiate into plasma cells which produce antibodies. (1)
The antibodies cause viruses to clump together (1)
This prevents the viruses from entering body cells. (1)
(Or other reasonable answers)

(c) The inhibitor greatly reduces the activity of neuraminidase. (1)
The viruses would remain bound to the host cells and could not leave to infect other cells. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)(i)	Understanding basic concepts	3
(b)(ii)	Understanding basic concepts	3
(c)	Applying concepts	2

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|!|BCELB0250000015|!| ★ **Cross topic**

Read the passage below and answer the questions that follow.

AIDS

AIDS is caused by the human immunodeficiency virus (HIV). The virus attacks certain cells that are involved in body defence. When people are infected by HIV, the lymph glands throughout their bodies become swollen. They lose weight and develop fever or even pneumonia. The mortality rate of AIDS is high. Many AIDS patients died of pneumonia.

HIV can be found in body fluids such as the blood, semen and vaginal secretions. Contact with body fluids from an infected person can transmit this virus.

(a) The cells that are attacked by HIV can activate B cells and phagocytes.

(i) What is the cell type that is attacked by HIV? (1 mark)

(ii) Describe how these cells are formed. (3 marks)

(b) Explain how AIDS is caused by HIV. (3 marks)

(c) Suggest how HIV may be transmitted from an HIV-infected mother to her infant. (1 mark)

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(d) Suggest ***two*** ways to prevent the transmission of HIV. (2 marks)

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_____

##

(a) (i) helper T cells (1)

(ii) T cells are produced in the bone marrow (1)
and then mature in the thymus gland. (1)
When T cells recognize the antigens present on the surfaces of infected cells, they are activated and differentiate into helper T cells. (1)

(b) HIV infects and destroy helper T cells, and leads to a drop in the number of helper T cells. (1)
The level of helper T cells in the patient's body drops to a low level and this weakens the patient's immunity. (1)
The patient becomes more susceptible to common infections (e.g. pneumonia). (1)

(c) Any ***one*** of the following: (1)
- HIV can pass through the placenta and enter the foetus's blood.
- The infant may be exposed to the mother's blood during labour.
- HIV may be found in the mother's milk and the infant could be infected during breastfeeding.

(d) Any ***two*** of the following: (2)
- Wear a condom during sexual intercourse
- Screen blood before transfusion
- Avoid sharing syringe needles

(Or other reasonable answers)

_____
(10 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	3
(b)	Applying concepts	3
(c)	Applying concepts	1
(d)	Understanding basic concepts	2

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|!|BCELB0250000016|!| ★ **Cross topic**

Rabies is an acute infection of the central nervous system caused by the rabies virus. It affects mammals such as dogs, bats and humans. When the skin of humans are bitten or scratched by an infected animal, the virus in the saliva or the skin of the infected animal enters the human body through the wound and travels through nerves to the brain, leading to encephalitis (inflammation of the brain).

Japanese encephalitis is caused by the Japanese encephalitis virus and is transmitted by the bite of infected mosquitoes called *Culex tritaeniorhynchus*. The incidence of the disease is higher in the rural and agricultural areas in Asia and the Western Pacific Region. It is a rare disease in Hong Kong.

(a) Give ***two*** similarities between rabies and Japanese encephalitis. (2 marks)

(b) Vaccination can prevent rabies and Japanese encephalitis.
How can the injection of vaccines prevent people from contracting these diseases? (3 marks)

(c) Apart from vaccination, rabies and Japanese encephalitis can be prevented by other ways. Suggest ***two*** ways for each disease. (4 marks)

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##

(a) Any ***two*** of the following: (2)

- Both diseases can cause encephalitis / cause harm to the central nervous system in humans.
- Both diseases are caused by viruses.
- Both diseases are transmitted by vectors (i.e. dogs for rabies and mosquitoes for Japanese encephalitis)

(Or other reasonable answers)

(b) After vaccination, antigens from the vaccine stimulate the primary response. (1)

Afterwards, the antigen of the virus is memorized by memory cells. (1)

Subsequent exposure to the antigens of the same virus results in the rapid secondary response, leading to the production of a large amount of antibodies against the virus within a short time. (1)

(c) Any ***two*** of the following prevention against rabies: (2)

- Vaccinate pet dogs and stray dogs against rabies
- Avoid contact stray animals, including dogs, cats and monkeys
- Wash any wounds immediately with disinfectants after being bitten or scratched

(Or other reasonable answers)

Any ***two*** of the following prevention against Japanese encephalitis: (2)

- Drain away stagnant water, which is the breeding site of mosquito larvae
- Kill mosquito larvae by spraying larvicidal oil in water bodies
- Kill adult mosquitoes with insecticides
- Apply insect repellents to exposed areas of the skin

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	3
(c)	Understanding basic concepts	4

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|!|BCELB0250000017|!| ★ **Cross topic**

Both antibiotics and vaccines can be used to stop the transmission of infectious diseases.

(a) Explain how antibiotics and vaccines can stop the transmission of infectious diseases. (4 marks)

(b) State ***two*** differences between antibiotics and vaccines in the control of infectious diseases. (2 marks)

(c) Give ***one*** problem associated with the use of antibiotics and vaccines, respectively. (2 marks)

##

(a) Antibiotics can be used to treat infectious diseases by interrupting the metabolism of pathogens. (1)
This can stop the transmission of disease from a cured patient to other people. (1)
Vaccination can induce immunity against infectious diseases. (1)
When the majority of the population is immunized against a certain disease, the disease will not spread easily in the community. (1)

(b) Antibiotics are taken after infections to treat infections while vaccines are injected before infections to prevent infections. (1)
Antibiotics target on bacteria while vaccines target on both bacteria and viruses. (1)
(Or other reasonable answers)

(c) Antibiotics:
Misuse of antibiotics may speed up the development of antibiotic resistance in bacteria. (1)

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Vaccines:

Side effects such as a low-grade fever, or pain and redness at injection site may appear / Some people may be allergic to certain components in vaccines. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Applying concepts	2
(c)	Applying concepts	2

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